

C613

Middle School Mathematics: Content Knowledge

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A baker who has run a shop downtown since 2010 uses 3 cups of flour for 4 dozen cookies. The shop's best seller is a lemon tart. How many cups of flour are needed for 10 dozen cookies?

- A. 12 cups
- B. About 13.3 cups
- C. 7.5 cups
- D. 9 cups

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A store manager with 12 years in retail raises a \$80 jacket's price by 25%, then later marks it down 20%. The jacket comes in four colors. What is the final price?

- A. \$100
- B. \$85
- C. \$80
- D. \$84

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A jacket sells for \$60 after a 25% discount. What was the original price?

- A. \$75
- B. \$80
- C. \$85
- D. \$45

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A middle-school art teacher with a kiln in her room builds a cube model with 2 cm edges, then a similar cube with 6 cm edges. The class meets on Tuesdays and Thursdays. How many times as much paint does the larger cube's surface need?

- A. 9 times
- B. 27 times
- C. 6 times
- D. 3 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Two similar water tanks: the larger holds 8 times the volume of the smaller. What is the ratio of their heights?

- A. 4 to 1
- B. 8 to 1
- C. 2 to 1
- D. About 2.83 to 1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A painter whose company has five trucks leans a 13-foot ladder against a wall, with its base 5 feet from the wall on level ground. The job is on a two-story house. How high up the wall does the ladder reach?

- A. 8 feet
- B. About 13.9 feet
- C. 18 feet
- D. 12 feet

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A small nonprofit with a volunteer board reports five staff salaries: \$40,000, \$42,000, \$45,000, \$48,000, and \$250,000. The office is in a converted warehouse. Which value best describes a typical salary?

- A. The mean, \$85,000
- B. The range, \$210,000
- C. The mode
- D. The median, \$45,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A very large value is added to a data set. How do the mean and median typically respond?

- A. The mean rises a lot; the median changes little.
- B. Both change by exactly the same amount.
- C. The median rises a lot; the mean changes little.
- D. Only the mode changes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A game at a school carnival that raised \$2,000 last year uses a bag with 5 red and 3 blue marbles. A player draws two marbles without replacement. The prize is a stuffed bear. What is the probability both are red?

- A. $\frac{5}{14}$
- B. $\frac{5}{8} + \frac{4}{7}$
- C. $\frac{25}{64}$
- D. $\frac{5}{8}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A fair coin is flipped and a fair number cube is rolled. What is the probability of heads and a number greater than 4?

- A. $\frac{1}{12}$
- B. $\frac{1}{6}$
- C. $\frac{5}{6}$
- D. $\frac{1}{3}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A carpenter who restores old homes has $\frac{3}{4}$ yard of trim and cuts pieces $\frac{1}{8}$ yard long. She bought the trim at a lumberyard that closes at 5 p.m. How many pieces can she cut?

- A. $\frac{1}{6}$
- B. 6
- C. $\frac{5}{8}$
- D. $\frac{3}{32}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A gym near a busy highway charges a \$25 sign-up fee plus \$10 per month. The gym has 300 members. Is the total cost proportional to the number of months?

- A. Yes, with a constant of 10.
- B. Yes; every linear relationship is proportional.
- C. Yes, with a constant of 35.
- D. No; the graph does not pass through the origin, so $\text{cost} \div \text{months}$ is not constant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A baker ~~who has run a shop downtown since 2010~~ uses 3 cups of flour for 4 dozen cookies. ~~The shop's best seller is a lemon tart.~~ How many cups of flour are needed for 10 dozen cookies?

Target word: flour for 10 dozen **Decides:** $\frac{3}{4}$ cup per dozen $\times 10$.

A	Wrong stem number	Multiplies the stem numbers 3×4 .
B	Inverted operation	Inverts the ratio: $4 \times 10 \div 3$.
C	Correct	$3 \div 4 = 0.75$ cup per dozen; $0.75 \times 10 = 7.5$.
D	Adjacent category	Additive reasoning: 6 more dozen, so 6 more cups.

2. Answer C Concept 2

A store manager ~~with 12 years in retail~~ raises a \$80 jacket's price by 25%, then later marks it down 20%. ~~The jacket comes in four colors.~~ What is the final price?

Target word: final price after two changes **Decides:** Multiply the factors: $80 \times 1.25 \times 0.80$.

A	Dropped element	Drops the markdown.
B	Adjacent category	Treats the percents as dollars: $80 + 25 - 20$.
C	Correct	$80 \times 1.25 = 100$; $100 \times 0.80 = 80$.
D	Confusion set	Adds the percents ($+25 - 20 = +5\%$).

3. Answer B Concept 2 · transfer

A jacket sells for \$60 after a 25% discount. What was the original price?

Target word: solve for the original (base) **Decides:** Sale = $0.75 \times$ original.

A	Inverted operation	Adds 25% to the sale price; the discount was on the original.
B	Correct	$60 \div 0.75 = 80$.
C	Adjacent category	Adds 25 dollars instead of 25%.
D	Confusion set	Takes another 25% off the sale price.

4. Answer A Concept 3

A middle-school art teacher ~~with a kiln in her room~~ builds a cube model with 2 cm edges, then a similar cube with 6 cm edges. ~~The class meets on Tuesdays and Thursdays.~~ How many times as much paint does the larger cube's surface need?

Target word: surface-area ratio **Decides:** Scale factor 3; area scales by $3^2 = 9$.

A	Correct	Area scales by $k^2 = 3^2 = 9$.
B	Confusion set	That is the volume ratio (k^3).
C	Wrong stem number	Uses the larger edge length as the ratio.
D	Dropped element	Uses the length scale factor for an area.

5. Answer C Concept 3 • transfer

Two similar water tanks: the larger holds 8 times the volume of the smaller. What is the ratio of their heights?

Target word: reverse: length ratio from volume ratio **Decides:** $k^3 = 8$.

A	Confusion set	Uses the area relationship.
B	Dropped element	Treats the volume ratio as a length ratio.
C	Correct	The cube root of 8 is 2.
D	Inverted operation	Takes a square root instead of a cube root.

6. Answer D Concept 4

A painter ~~whose company has five trucks~~ leans a 13-foot ladder against a wall, with its base 5 feet from the wall on level ground. ~~The job is on a two-story house.~~ How high up the wall does the ladder reach?

Target word: missing leg **Decides:** $13^2 - 5^2 = 144$.

A	Wrong stem number	Subtracts the lengths: $13 - 5$.
B	Confusion set	Treats the ladder as a leg: $\sqrt{169 + 25}$.
C	Inverted operation	Adds the lengths instead of using squares.
D	Correct	$\sqrt{169 - 25} = \sqrt{144} = 12$.

7. Answer D Concept 5

A small nonprofit ~~with a volunteer board~~ reports five staff salaries: \$40,000, \$42,000, \$45,000, \$48,000, and \$250,000. ~~The office is in a converted warehouse.~~ Which value best describes a typical salary?

Target word: typical value in skewed data **Decides:** One large outlier.

A	Confusion set	The outlier pulls the mean far above four of five salaries.
B	Wrong stem number	Range measures spread, not a typical value.
C	Adjacent category	No value repeats, so there is no mode.
D	Correct	The median resists the \$250,000 outlier.

8. Answer A Concept 5 • transfer

A very large value is added to a data set. How do the mean and median typically respond?

Target word: reverse: effect of an outlier **Decides:** The mean uses every value's size.

A	Correct	The median depends only on position.
B	Absolute language	They respond very differently to outliers.
C	Inverted operation	Reverses which measure is sensitive.
D	Adjacent category	A single new value rarely changes the mode.

9. Answer A Concept 6

A game at a school carnival ~~that raised \$2,000 last year~~ uses a bag with 5 red and 3 blue marbles. A player draws two marbles without replacement. ~~The prize is a stuffed bear.~~ What is the probability both are red?

Target word: P(both red), no replacement **Decides:** $5/8 \times 4/7$.

A	Correct	$5/8 \times 4/7 = 20/56 = 5/14$.
B	Inverted operation	Adds instead of multiplying; the total exceeds 1.
C	Dropped element	Drops "without replacement": $5/8 \times 5/8$.
D	Wrong stem number	Only the first draw.

10. Answer B Concept 6 · transfer

A fair coin is flipped and a fair number cube is rolled. What is the probability of heads and a number greater than 4?

Target word: independent "and" **Decides:** $1/2 \times 2/6$.

A	Wrong stem number	Counts only one number (6) above 4.
B	Correct	$1/2 \times 1/3 = 1/6$.
C	Inverted operation	Adds $1/2 + 1/3$.
D	Dropped element	Drops the coin.

11. Answer B Concept 7

A carpenter ~~who restores old homes~~ has $3/4$ yard of trim and cuts pieces $1/8$ yard long. She bought the trim ~~at a lumberyard that closes at 5 p.m.~~ How many pieces can she cut?

Target word: how many groups fit **Decides:** $3/4 \div 1/8$.

A	Confusion set	Flips the dividend instead of the divisor.
B	Correct	$3/4 \times 8/1 = 6$.
C	Adjacent category	Subtracts one piece instead of dividing.
D	Inverted operation	Multiplies instead of dividing.

12. Answer D Concept 8

A gym ~~near a busy highway~~ charges a \$25 sign-up fee plus \$10 per month. ~~The gym has 300 members.~~ Is the total cost proportional to the number of months?

Target word: proportional or not **Decides:** The fee means cost $\neq 0$ at 0 months.

A	Dropped element	Drops the \$25 fee.
B	Absolute language	Only lines through the origin are proportional.
C	Wrong stem number	Adds the fee and rate into one constant.
D	Correct	$y = 10x + 25$ is linear but not proportional.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	7	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Proportional reasoning	Equivalent ratios scale multiplicatively: find the unit rate or scale factor and multiply.	Adding the same amount to both quantities (additive reasoning).	A recipe, map, or rate scaled to a new amount.
2	Percent change uses the original as the base	Percent change = change ÷ original; successive percent changes multiply $(1 + r_1)(1 + r_2)$ and do not simply add.	Adding and subtracting percents, or dividing by the new value.	Two percent changes in a row, or finding an original price.
3	Scale factor effects	If lengths scale by k , areas scale by k^2 and volumes by k^3 .	Scaling area or volume by k .	"Similar" figures or a scale model with paint, fabric, or capacity.
4	Pythagorean theorem	In a right triangle $a^2 + b^2 = c^2$, where c is the hypotenuse opposite the right angle.	Adding or subtracting side lengths, or treating a leg as the hypotenuse.	A ladder, diagonal, or distance between points.
5	Mean vs. median with outliers	The mean is pulled toward outliers; the median resists them and better shows a typical value in skewed data.	Always reporting the mean as the typical value.	One value far from the rest.
6	Compound probability	For independent events multiply probabilities; without replacement, the second probability changes.	Adding probabilities for "and", or ignoring that items are not replaced.	"Both", "and", or "without replacement".
7	Dividing fractions	Dividing by a fraction means multiplying by its reciprocal; it answers "how many groups of this size fit".	Multiplying straight across, or flipping the wrong fraction.	"How many pieces of length ... can be cut?"
8	Proportional vs. non-proportional linear	A relationship is proportional only if $y = kx$: the ratio y/x is constant and the graph passes through the origin.	Calling every straight-line relationship proportional.	A start-up fee or flat charge plus a rate.

Course scope (public description): Mathematics: Middle School Content Knowledge is designed to help candidates refine and integrate the mathematics content knowledge and skills necessary to become successful middle school mathematics teachers. A high level of mathematical reasoning skills and the ability to solve problems are necessary to complete this course. Prerequisites for this course are College Geometry, Probability and Statistics I, and Pre-Calculus.